

Stablecoin Stability Analysis: Evaluating Peg Deviations in Major Cryptocurrencies

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Introduction to Data Analysis and Visualization

October 13, 2025

Why Stablecoin Stability Matters I

What are Stablecoins?

- Digital currencies designed to maintain \$1.00 value
- Bridge between crypto and traditional finance
- Market size: \$300+ billion (Oct 2025)

Recent Crisis Event:

- **October 10-11, 2025:** USDe crashed from \$1.00 to \$0.65
- **\$19.37 billion liquidated** in 24 hours
- Largest liquidation event in crypto history

Historical Depeg Events:

- **Terra/Luna (May 2022):** \$50B lost
- **USDC (SVB Crisis, March 2023):** Dropped to \$0.87
- **TUSD (January 2024):** Fell to \$0.926

Project Goal:

- Build analytical tools to detect instability early
- Understand what causes depegging
- Compare stability across different stablecoins

Why Stablecoin Stability Matters II

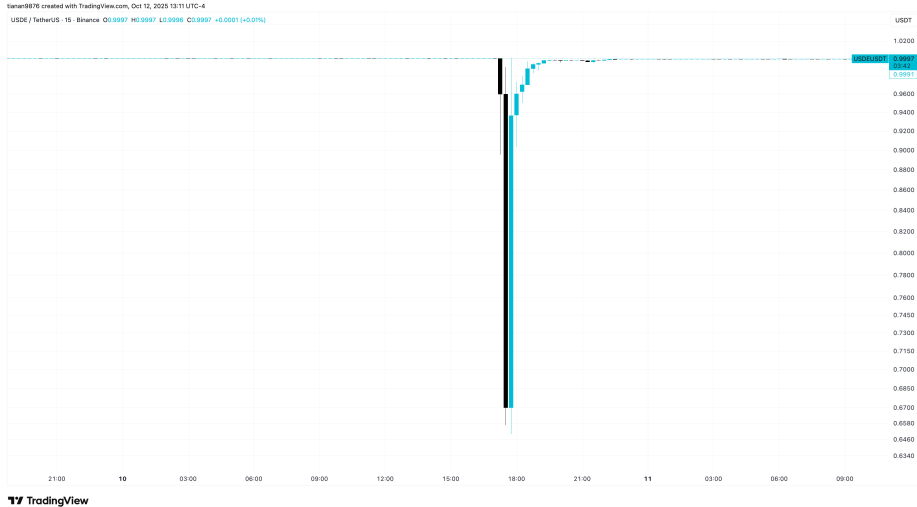


Figure: USDe Depeg Event, October 10-11, 2025

Dataset Overview

Stablecoins Analyzed (6-8):

- **USDT** (Tether) - Largest, \$95B+ cap
- **USDC** (USD Coin) - Most regulated
- **DAI** - Decentralized
- **BUSD** (Binance USD)
- **TUSD, FRAX, USDP, GUSD**

Data Source:

- CoinGecko API
- Binance API
- Public blockchain data

Dataset Specifications:

Parameter	Value
Time Period	18-24 months
Granularity	Hourly data
Total Samples	60,000-80,000
Stablecoins	6-8
Features	20+
Classes	2-3 (stability levels)

Key Features (20+):

- Price deviation from \$1.00 (abs & %)
- Trading volume (hourly/daily/weekly)
- Volatility metrics (rolling std dev)
- Temporal features (hour, day, week)
- Market cap changes
- Cross-correlation between coins

Method 1: Clustering Analysis

Objective: Group time periods by similar stability characteristics

Approach:

- **K-Means Clustering**
- Identify distinct stability regimes
- Cluster into 3 groups:
 - High Stability
 - Medium Stability
 - Low Stability (Crisis)

Features Used:

- Deviation magnitude
- Deviation duration
- Recovery time
- Trading volume
- Price volatility

Expected Insights:

- Identify periods with similar behavior patterns
- Detect "crisis clusters" across multiple coins
- Analyze contagion effects between stablecoins
- Compare stability profiles

Applications:

- Market stress identification
- Risk categorization
- Portfolio diversification strategies

Method 2: Classification Models

Objective: Predict stability levels based on current market features

Algorithms:

1 Decision Trees

- Interpretable rules
- Clear feature importance
- Visual decision paths

2 Random Forest

- Ensemble method
- Higher accuracy
- Handles non-linearity
- Robust to overfitting

Target Classes:

- High Stability (deviation smaller than 0.1%)
- Medium Stability (0.1% - 1%)
- Low Stability (bigger than 1%)

Key Features:

- Rolling price averages (1h, 24h, 7d)
- Volume trends and changes
- Historical deviation patterns
- Volatility indicators
- Bitcoin/Ethereum price movements

Expected Results:

- Feature importance ranking
- Accuracy: 80-90% (expected)
- Early warning capability (1-6 hours ahead)
- Identify most predictive indicators

Evaluation Metrics:

- Accuracy, Precision, Recall
- F1-Score, Confusion Matrix

Method 3: Anomaly Detection I

Objective: Automatically identify significant de-pegging events

Approach:

- **Isolation Forest Algorithm**
- Unsupervised learning method
- Detects outliers in deviation patterns
- Dynamic threshold adjustment

Anomaly Criteria:

- Deviation is bigger than 2% lasting more than 1 hour
- Sudden volume spikes (bigger than 3σ)
- Rapid price movements
- Correlation breakdowns

Detection Features:

- Deviation magnitude and velocity
- Volume anomalies
- Price volatility spikes
- Multi-coin correlation patterns

Real-World Example:

- **October 10, 2025: USDe**
- Deviation: 35% (\$1.00 → \$0.65)
- Duration: 2-3 hours
- **Model flags as high-severity anomaly**

Method 3: Anomaly Detection II

Objective: Automatically identify significant de-pegging events

Applications:

- Real-time monitoring systems
- Automated alert generation
- Risk management tools
- Early warning for traders/investors

Performance Metrics:

- True Positive Rate (sensitivity)
- False Positive Rate
- Detection latency

Method 4: Regression Analysis

Objective: Predict magnitude of deviation from \$1.00 peg

Algorithms:

- **Linear Regression** (baseline)
- **Ridge Regression** (L2 regularization)
- Predict continuous deviation values

Target Variable:

- Deviation percentage in next time window
- Range: -5% to +5%
- Time horizons: 1h, 6h, 24h

Input Features:

- Volume changes (hourly/daily)
- Price volatility metrics
- Market cap fluctuations
- Bitcoin/Ethereum market conditions
- Historical deviation trends
- Order book depth (if available)

Key Research Questions:

- 1 What factors correlate with larger deviations?
- 2 Can we predict recovery time?
- 3 How do different stablecoin types differ?
- 4 What role does overall market stress play?

Expected Findings:

- Volume spikes predict larger deviations
- Decentralized (DAI) vs. centralized (USDC) behavior differs
- Bitcoin drops increase deviation risk
- Market cap size affects stability

Evaluation Metrics:

- R^2 Score, MAE, RMSE

Implementation Roadmap I

Week-by-Week Project Plan:

Week	Period	Tasks
1	Oct 14-20	Data Collection & EDA Fetch data, EDA, identify depeg events, pre-processing
2	Oct 21-27	Methods 1 & 2 Implementation K-means clustering, Decision Tree & Random Forest
3	Oct 28-Nov 3	Methods 3 & 4 Implementation Isolation Forest, Linear/Ridge regression
4	Nov 4-10	Model Evaluation & Comparison Performance metrics, feature importance, cross-validation
5	Nov 11-17	Visualization & Final Report Dashboards, documentation, presentation

Implementation Roadmap II

Week-by-Week Project Plan:

Deliverables:

- Complete Python code (Jupyter notebooks)
- Interactive visualizations (Plotly/Matplotlib)
- Final presentation & technical report
- GitHub repository with documentation

Expected Visualizations & Insights

Visualization Types:

- ① **Time Series:** Price deviation, volume patterns, multi-coin comparison
- ② **Statistical Plots:** Distributions, box plots, correlation heatmaps
- ③ **Model Results:** Clusters (2D/3D), decision trees, feature importance, confusion matrices
- ④ **Anomaly Detection:** Highlighted depeg events, alert timeline

Key Insights Expected:

- **Stability Ranking:** Which stablecoin is most stable?
- **Contagion Effect:** Do depegs spread between coins?
- **Predictive Factors:** What signals warn of instability?
- **Recovery Patterns:** How quickly do coins restore peg?
- **Market Conditions:** Impact of BTC/ETH on stability

Interactive Dashboards:

- Real-time monitoring interface
- Historical analysis explorer
- Stablecoin comparison tool
- Risk assessment calculator

Business Value:

- Risk management for crypto investors
- Early warning system development
- Regulatory compliance insights
- Portfolio optimization guidance

Project Summary & Expected Contributions

Technical Contributions:

- Multi-method analysis framework
- 4 approaches: Clustering, Classification, Anomaly Detection, Regression
- Large-scale dataset (600K-800K samples)
- Real-world validation (Oct 2025 crash)

Practical Impact:

- Help investors assess stablecoin risks
- Enable early warning systems
- Support regulatory oversight
- Improve market transparency

Novel Aspects:

- Multi-stablecoin comparative analysis
- Hourly granularity (vs. daily in literature)
- Recent data including 2025 events
- Integrated multi-method approach

Success Criteria:

- Classification accuracy bigger than 80%
- Anomaly detection with low false positives
- Clear feature importance insights
- Actionable recommendations

Future Extensions:

- Real-time deployment
- Deep learning models (LSTM)
- Sentiment analysis integration
- Mobile alert application

Thank You!

Questions?

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